Transparent Saliency Routing for Temporal Image Drift

A compact mock research paper • fixed-horizon prototype run • 5 October 2026

Abstract

This prototype evaluates a ViT-only drift correction pipeline whose update decision can be inspected directly. Historical models train on 2004–2008, model selection uses 2009, drift adaptation uses 2010, router calibration uses 2011, and 2012–2014 remain untouched until final evaluation. The enhanced model reaches 91.99% macro F1 under 2014 combined drift and transparent routing reaches 91.85%. A matched five-epoch ablation attributes +0.13 points always-new and +0.23 points routed to the explicit auxiliary saliency objectives; the predeclared 3-point target remains unmet.

1. Research question

Can visible saliency change and prediction evidence determine when a drift-adapted ViT should replace a historical ViT prediction across a realistic future horizon?

2. Architecture and transparent routing

The experiment compares a baseline ViT, an attention-regularised XAI ViT, and a paired historical/newer XAI system. Its router is a standardized logistic rule over four logged quantities: saliency Jensen–Shannon divergence, confidence gain, entropy reduction, and class-evidence gain. A minimum saliency-change floor must also be passed. The saved coefficients and threshold make every update decision reproducible.

2.1 Model mechanics

DINOv2-small emits a class token and patch-token grid. The baseline pools patches uniformly. The XAI model learns one gate per patch, normalises the gates into a saliency distribution, and concatenates the saliency-weighted patch sum with the class token before classification. The explanation therefore participates directly in prediction.

Component	Mechanism	Role in drift correction
DINOv2-small	Class token + patch tokens	Shared ViT representation
Baseline head	Uniform patch pooling	Historical reference
XAI saliency head	Learned scalar gate per patch	Weights patches inside forward pass
Counterfactual path	Complement of salient patches	Tests class-evidence faithfulness
Twin comparison	Frozen historical + adapted newer ViT	Measures prediction and map change
Transparent router	Four-feature logistic rule	Chooses old or new prediction

2.2 Saliency faithfulness and update loss

A complementary path pools low-saliency patches. Training penalises the model unless selected patches provide greater true-class evidence than their complement. During adaptation, the pairwise term keeps historical and newer maps close when the historical prediction is correct, while imposing a divergence margin when it is wrong. Classification, pairwise saliency, and counterfactual evidence losses are optimized jointly.

2.3 Routing equation

For saliency JS divergence, confidence gain, entropy reduction, and evidence gain, the router computes z = (f – mean) / scale and p(update) = sigmoid(w·z + b). It selects the newer class only when the models disagree, saliency JS clears the saved floor, and p(update) clears the saved threshold. Every term is logged for sample-level audit.

3. Chronological protocol

The split models deployment: historical learning stops before adaptation, and the future horizon is used only for final measurement. This prevents year-by-year retraining from leaking test information into the model.

Stage	Years	Purpose
Historical training	2004–08	Fit original ViTs
Selection	2009	Choose checkpoints
Adaptation	2010	Update XAI ViT with mild drift
Calibration	2011	Fit explicit route rule
Frozen test	2012–14	Measure future drift

3.1 Optional realistic drift

The drift toggle applies mild saturation change, Gaussian sensor noise, and partial occlusion. The newer ViT sees a deterministic 50/50 mixture of clean and individually corrupted 2010 samples. The router is calibrated on moderate individual corruptions in 2011. Combined corruption is withheld until 2012–2014, making it a composition shift rather than a memorised augmentation recipe.

3.2 Saved routing policy

Transparent feature	Coefficient
Saliency Js	+0.240
Confidence Gain	+0.440
Entropy Reduction	-0.321
Evidence Gain	-0.055
Route probability threshold	0.35
Saliency-change floor	0.0022

Coefficients act on standardized features. Their signs describe the 2011 calibration relationship; they are not causal effects. The explicit saliency floor ensures that saliency remains part of the decision rather than a decorative post-hoc plot.

4. Enhanced correction experiment

Status: completed as server run 20260929-enhanced-3pp. The 3-point value was fixed before inspection; the run did not meet that primary target.

Controlled item	Standard profile	Enhanced profile
2010 stream	50% clean; severity 1	25% clean; individual severity 1–3
Trainable backbone	Final ViT block	Final three ViT blocks
Learning rates	All trainable: 1e-5	Backbone 1e-5; heads 5e-5
Label smoothing	0.00	0.05
Saliency loss weight	0.10	0.25
Divergence margin	0.05	0.08
Saliency floor	Fixed median	2011-calibrated quantile

4.1 Fixed controls

Historical checkpoint, CLEAR-100 chronology, random seed, epoch budget, 2011 calibration labels, frozen 2012–2014 test streams, corruption code, batch size, and metrics must match the standard run. Combined corruption remains absent from adaptation and calibration.

4.2 Predeclared hypotheses

H1 not met: enhanced XAI gained 1.40 points over baseline ViT on 2014 combined drift. H2 met: routing retained 91.5% of the newer model's gain over historical XAI. H3 met: routed clean 2014 macro F1 was 0.06 points above historical XAI. Test labels must not be used for another tuning cycle.

4.3 Evaluation matrix

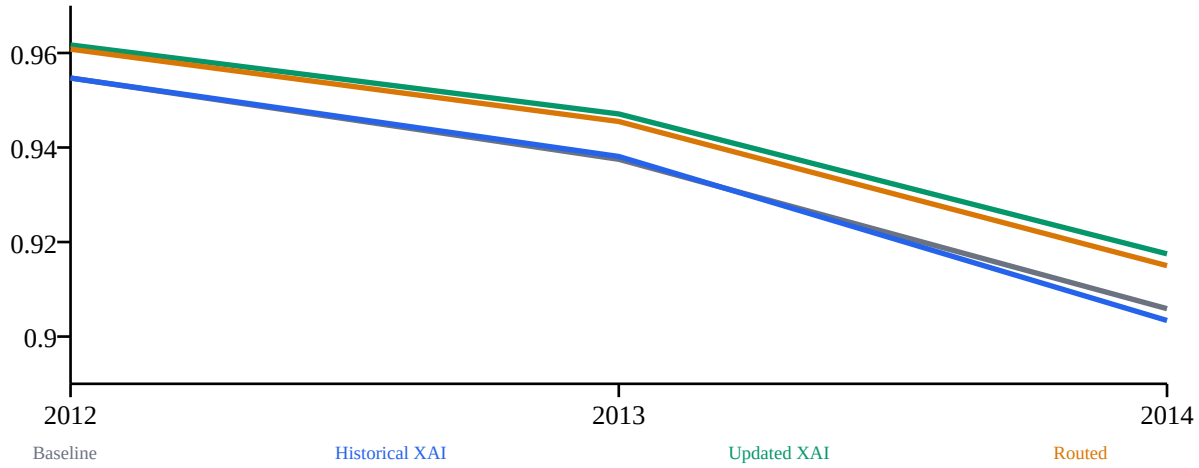
The selected epoch was 4, with 94.39% calibration macro F1; epoch 5 fell to 93.00%. Post-warm-up epochs took about 138 seconds. Final reporting includes accuracy, macro F1, macro IoU, update-trigger metrics, helpful and harmful routes, timing, and saliency-map overlap. Multiple seeds remain necessary before a final claim.

4.4 Checkpoint recovery test

Each evaluated epoch atomically writes model parameters, AdamW optimizer state, completed epoch, and best score. The matched standard run was deliberately interrupted after epoch 2 and resumed at epoch 3. Its optimizer state and best score were restored, epoch 4 was later selected, and the frozen evaluation completed. A local round-trip regression test separately verifies the serialized state and metadata.

5. Completed fixed-horizon results

Macro F1 under combined saturation, noise, and occlusion. The widening decline across years makes the temporal effect visible.



Year	Baseline	Hist. XAI	Updated XAI	Routed	Helpful / harmful
2012	95.47	95.47	96.17	96.08	45 / 14
2013	93.75	93.81	94.71	94.55	72 / 32
2014	90.59	90.34	91.75	91.50	120 / 48

In 2014, the updated XAI model gains 1.41 points over historical XAI. Routing retains 1.16 points: (91.50–90.34)/(91.75–90.34) = 82.3% of the available gain. Clean-data macro F1 is 95.13% baseline, 95.24% historical XAI, 95.85% updated XAI, and 95.46% routed.

5.1 Enhanced combined-drift result

Year	Hist. XAI	Std. updated	Enhanced updated	Enhanced routed	Helpful / harmful
2012	95.47	96.17	95.71	95.81	76 / 58
2013	93.81	94.71	94.43	94.44	101 / 66
2014	90.34	91.75	91.99	91.85	188 / 90

In 2014, enhanced XAI reaches 91.99%: +1.40 points over baseline ViT and +1.65 over historical XAI. Enhanced routing reaches 91.85%, retaining 91.5% of the available gain. Relative to the standard profile, enhancement adds 0.24 points always-new and 0.35 points routed. H1 fails; H2 and H3 pass.

5.2 Controlled five-epoch ablation

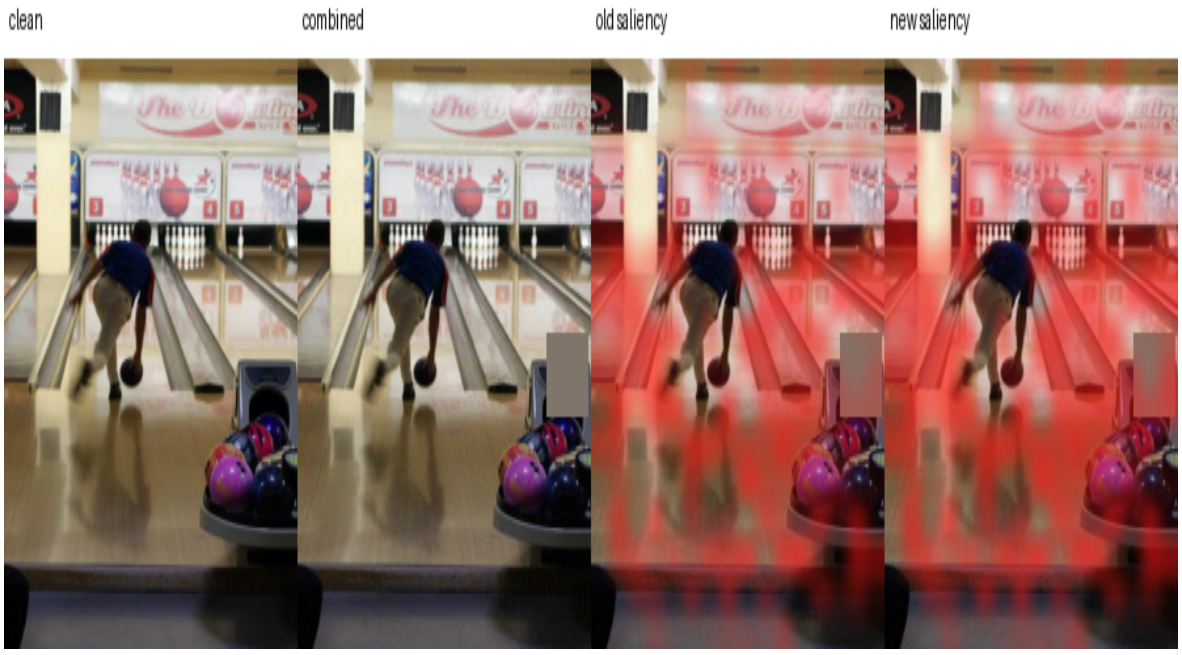
Five-epoch configuration	Clean routed	Combined new	Combined routed	Helpful / harmful
Matched standard	95.78	91.68	91.72	144 / 57
Enhanced + auxiliary XAI	95.30	91.99	91.85	188 / 90
Enhanced, no auxiliary XAI	95.45	91.86	91.62	125 / 43

The enhanced auxiliary saliency objectives changed 2014 combined macro F1 by +0.13 points for always-new inference and +0.23 points after routing. The control retains the saliency-gated forward pass and removes only the pairwise divergence and complementary-evidence objectives.

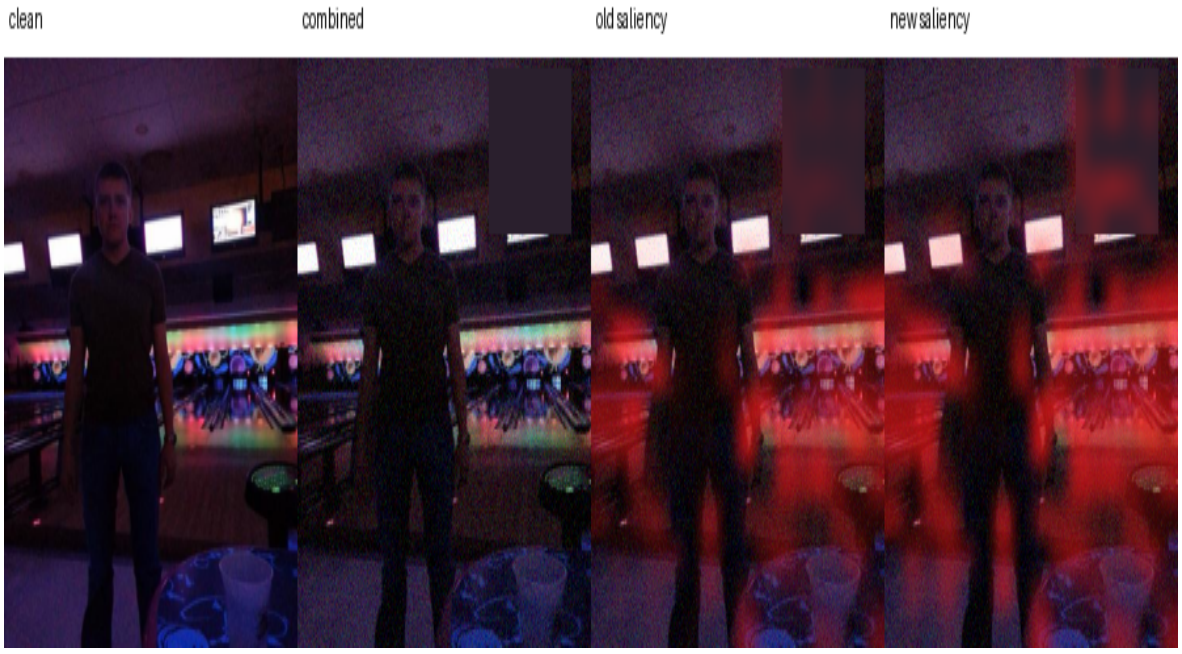
6. Saliency audit examples

Each panel shows the clean reference, drifted input, historical saliency, and updated saliency. Together with the saved decision log, this lets a reviewer connect the route to visible attention changes and prediction evidence.

Combined drift • 2012

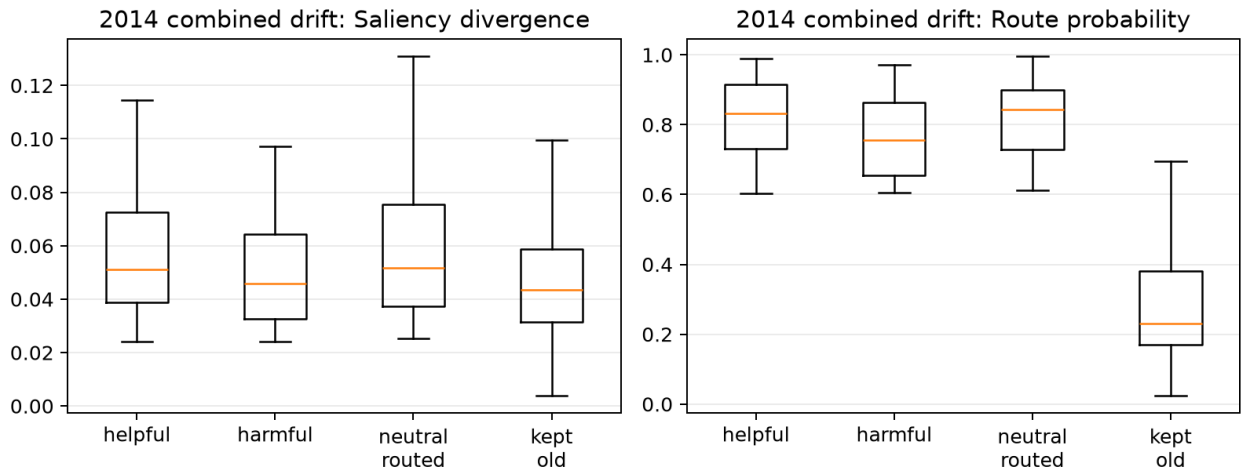


Combined drift • 2014



6.1 Saliency features by routing outcome

The box plots group the 2014 combined-drift decisions as helpful, harmful, neutral routed, or kept historical. They test whether visible saliency divergence and the explicit route probability separate useful from harmful updates; they do not establish that saliency caused the correction.



Helpful routes show mean saliency divergence 0.0607 versus 0.0512 for harmful routes and 0.0473 when the historical model is retained. Mean route probabilities are 0.821, 0.767, and 0.310, respectively. The overlap is substantial: visible change strongly separates routed from retained cases, while separating useful from harmful updates only modestly. Qualitative maps often emphasize contextual regions, so they are evidence gates rather than object masks.

7. Interpretation

Equalizing the five-epoch budget reduces the enhanced advantage over standard to +0.31 points always-new and +0.13 points routed on 2014 combined drift. Within the enhanced profile, explicit auxiliary saliency objectives add +0.13 points always-new and +0.23 points routed over the matched no-auxiliary-loss control. The gain is directionally positive but small. Full enhancement routes 388 samples with 188 helpful and 90 harmful switches; the control routes 245 with 125 helpful and 43 harmful switches.

8. Limitations and validity threats

All runs use one seed on one temporal stream with synthetic corruptions. Checkpoint selection and router calibration reuse 2011 examples, and there are no confidence intervals or external validation data. H1 is a valid negative result. The ablation isolates the explicit auxiliary saliency objectives, while the saliency gate remains inside both classifiers. Overlapping helpful and harmful outcome distributions preclude a strong causal explanation claim.

9. Next experimental sequence

Split 2011 into disjoint checkpoint-selection and router-calibration subsets, then repeat matched profiles across at least three seeds with paired per-sample bootstrap intervals. Add confidence-only and saliency-only routers. A stronger predeclared intervention should compare saliency-guided patch corruption with random patch corruption, followed by evaluation on a second temporal image dataset.

10. Reproducibility

The repository includes compact JSON comparisons, learned router policies, 90 new enhanced/control saliency panels, saliency outcome CSV/JSON/PNG artifacts, a checkpoint round-trip regression test, and the report generator. Large checkpoints remain server-side. New run identifiers are 20261005-standard-5ep and 20261005-enhanced-nosaliency; both select epoch 4.

References

[1] Dosovitskiy et al., Vision Transformer, 2020. [2] Oquab et al., DINOv2, 2023. [3] Hendrycks & Dietterich, ImageNet-C, 2019. [4] Hendrycks et al., AugMix, 2019. [5] Lin et al., CLEAR, 2021. Full links appear in the Markdown version.